

NAREDDY POOJITHA

NLP Engineer | Email: poojithanareddy@gmail.com | India

PROFESSIONAL SUMMARY

NLP Engineer with hands-on experience in building text classification, named entity recognition, question answering and text summarization systems using Python, PyTorch and Hugging Face Transformers. Skilled in fine-tuning BERT and other transformer models, building end-to-end NLP pipelines, and deploying models as REST APIs. Passionate about large language models, information extraction and semantic search.

TECHNICAL SKILLS

Programming: Python, SQL, Java (basic)

NLP Techniques: Tokenization, Lemmatization, POS Tagging, Named Entity Recognition (NER), Text Classification, Sentiment Analysis, Topic Modeling, Text Summarization, Word Embeddings, Semantic Search, TF-IDF

Libraries and Frameworks: NLTK, spaCy, Hugging Face Transformers, Gensim, Scikit-learn, PyTorch, TensorFlow, Keras, Pandas, NumPy

Models: BERT, RoBERTa, DistilBERT, GPT, T5, LSTM, Word2Vec, GloVe, FastText, LLMs

Tools and Deployment: Flask, FastAPI, Streamlit, Docker, Git, GitHub, Jupyter Notebook, AWS, MLflow

Databases: MySQL, MongoDB, FAISS, Pinecone

PROFESSIONAL EXPERIENCE

NLP Engineer Intern - Text Analytics Team

Jan 2025 - Jun 2025

- Built a text classification pipeline using DistilBERT to categorize customer support tickets, achieving 91% accuracy.
- Developed a Named Entity Recognition model with spaCy to extract names, dates and organizations from documents.
- Cleaned and preprocessed over 50,000 text records using NLTK and Pandas, improving model F1-score by 8%.
- Deployed the trained model as a REST API using FastAPI and Docker for real-time predictions.
- Created sentiment analysis dashboards with Streamlit for the product team.

PROJECTS

AI Resume Analyzer using NLP

- Built a web application that parses resumes, extracts skills using spaCy and matches them to job roles with a match score.
- Used TF-IDF and cosine similarity along with Sentence-BERT embeddings for semantic matching of skills.
- Technologies: Python, spaCy, Scikit-learn, Sentence Transformers, Streamlit.

Question Answering System using BERT

- Fine-tuned a BERT model on the SQuAD dataset to answer questions from a given context passage.
- Achieved an F1-score of 86% and built a simple Flask interface for testing.
- Technologies: Python, PyTorch, Hugging Face Transformers, Flask.

Text Summarization and Sentiment Analysis

- Implemented abstractive news summarization using the T5 transformer model and evaluated it with ROUGE scores.
- Trained an LSTM sentiment classifier on IMDB movie reviews with 89% accuracy.
- Technologies: Python, TensorFlow, Keras, NLTK, Hugging Face.

Semantic Search Engine using Embeddings and FAISS

- Built a document search engine using sentence embeddings and FAISS vector index for fast similarity search.
- Technologies: Python, Sentence Transformers, FAISS, FastAPI.

EDUCATION

B.Tech in Computer Science and Engineering (AI and ML)

2021 - 2025 | CGPA: 8.4 / 10

CERTIFICATIONS

- Natural Language Processing Specialization - Coursera (DeepLearning.AI)
- Hugging Face NLP Course - Hugging Face
- Machine Learning with Python - IBM

ACHIEVEMENTS

- Ranked in top 10 in a college-level NLP hackathon for building a fake news detection system.
- Published a technical article on fine-tuning BERT for text classification.